

Background

The **Rice Recreation Center** is often **crowded during peak hours**, especially in the **weight room**. To combat this, **staff attempt to manually monitor occupancy**.

Objective

Need
A **long-lasting, low-maintenance, automated** tracking system that **accurately reports weight room occupancy in hourly intervals** within a **10% margin of error**.

Impact
↓ Labor Costs
↑ Workout Efficiency
Reliable Data for facility expansions

Design Criteria

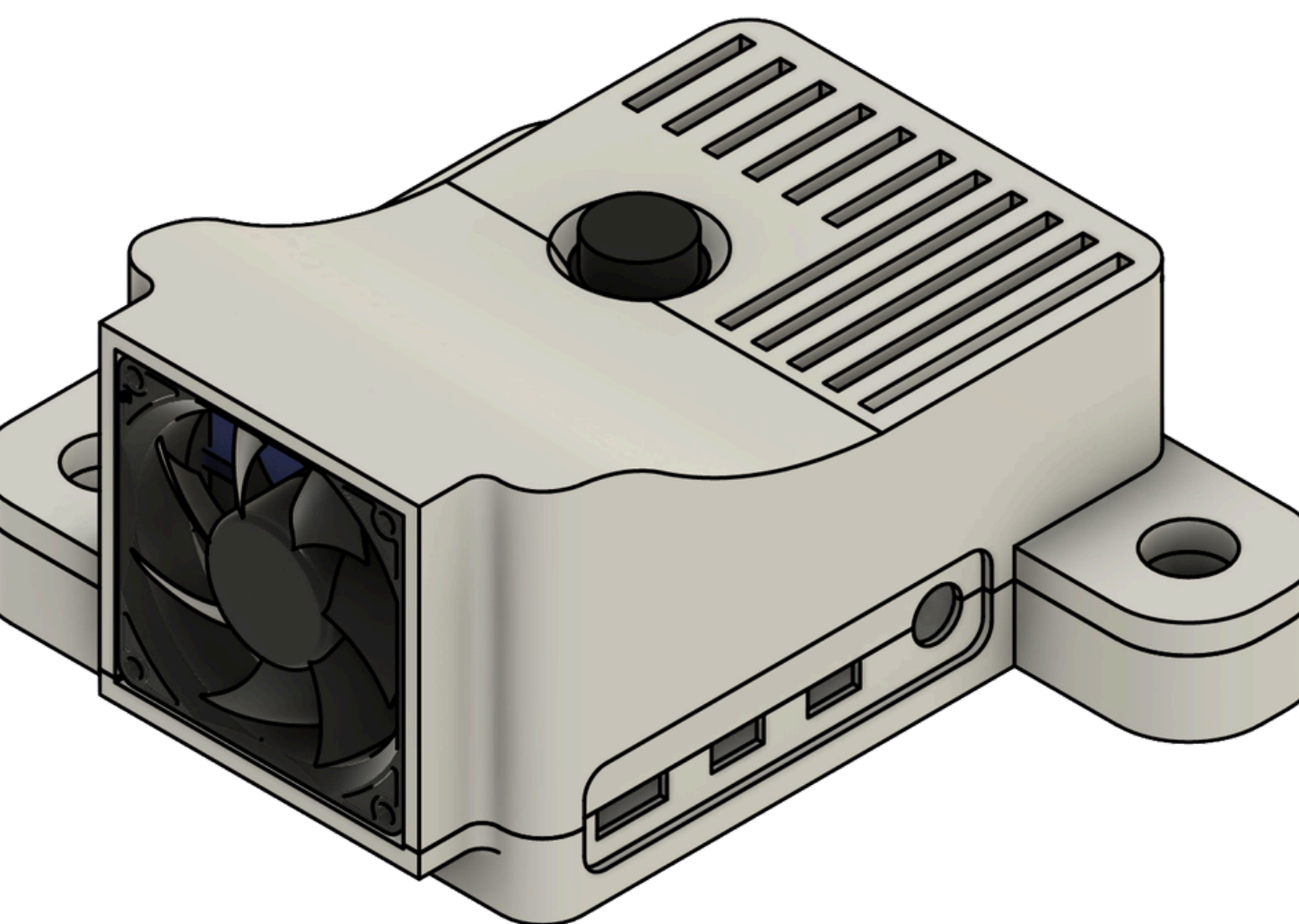
- Data Collection Frequency** ≥ 1 Update/Hour
- Confidentiality** No Biometric Data Stored
- Longevity** ≥ 10 Years
- Accuracy** ≥ 90%
- Cost (Per Device)** ≤ \$250
- User Intervention Frequency** User Intervenes < 2 Times Per Day
- Scalability** ≤ 1 Device / 100m²

Device Overview

Proposed Solution

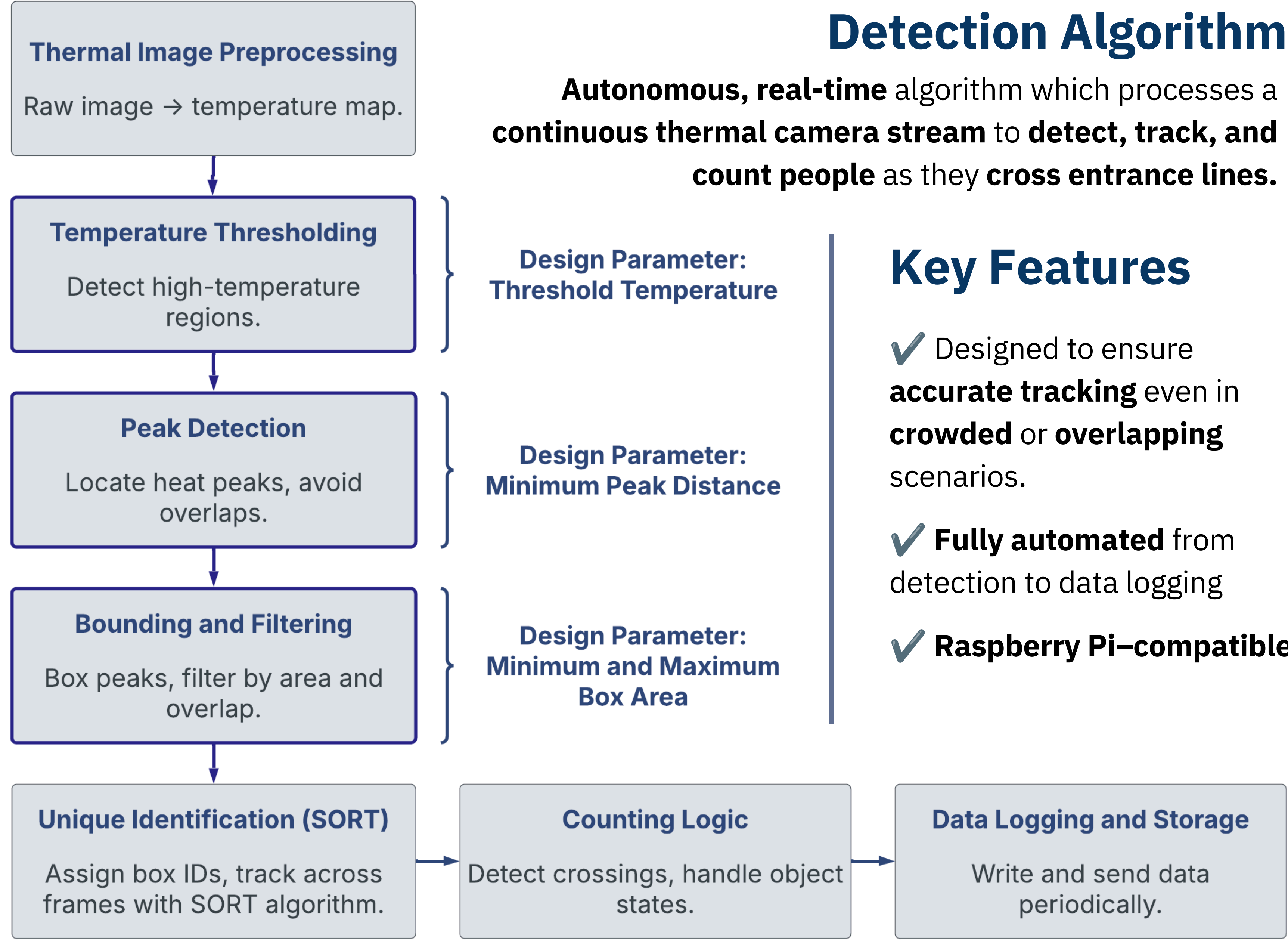
A **ceiling-mounted thermal camera** interfaced with a **Raspberry Pi** to **count people** leaving and entering the weight room through the **entrances**.

Assembly

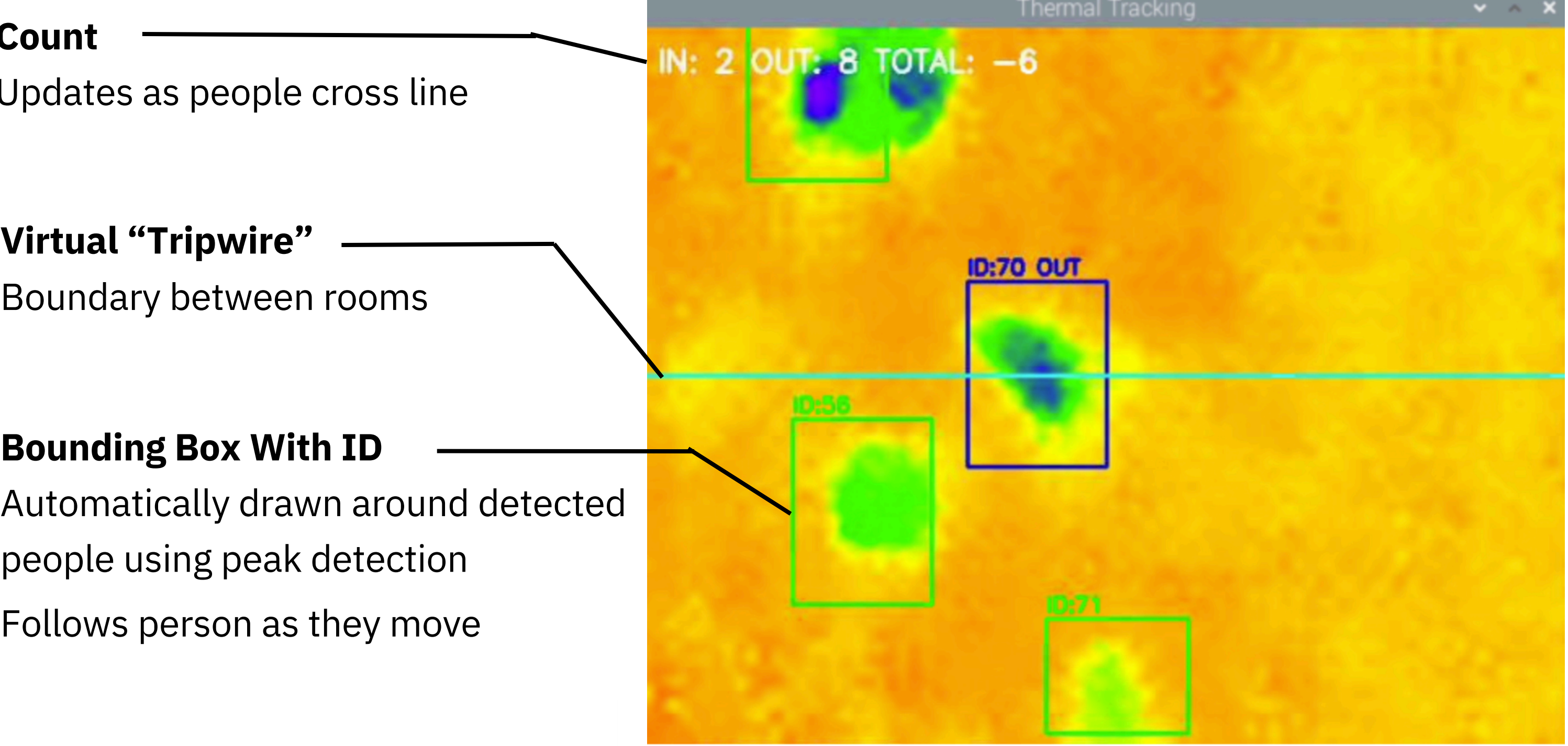


Component Breakdown

- 3D Printed Case Top**
Notched for Alignment
Air Intake Holes
- Cooling Fan**
Activates at 80°F
- IR Thermal Camera**
80x62 Pixel Resolution
25 FPS Max
- Raspberry Pi 4B**
4GB RAM
- 3D Printed Case Bottom**
Easy Port Access
PLA Plastic
Holes for Case, Platform for Command Strip Attachment

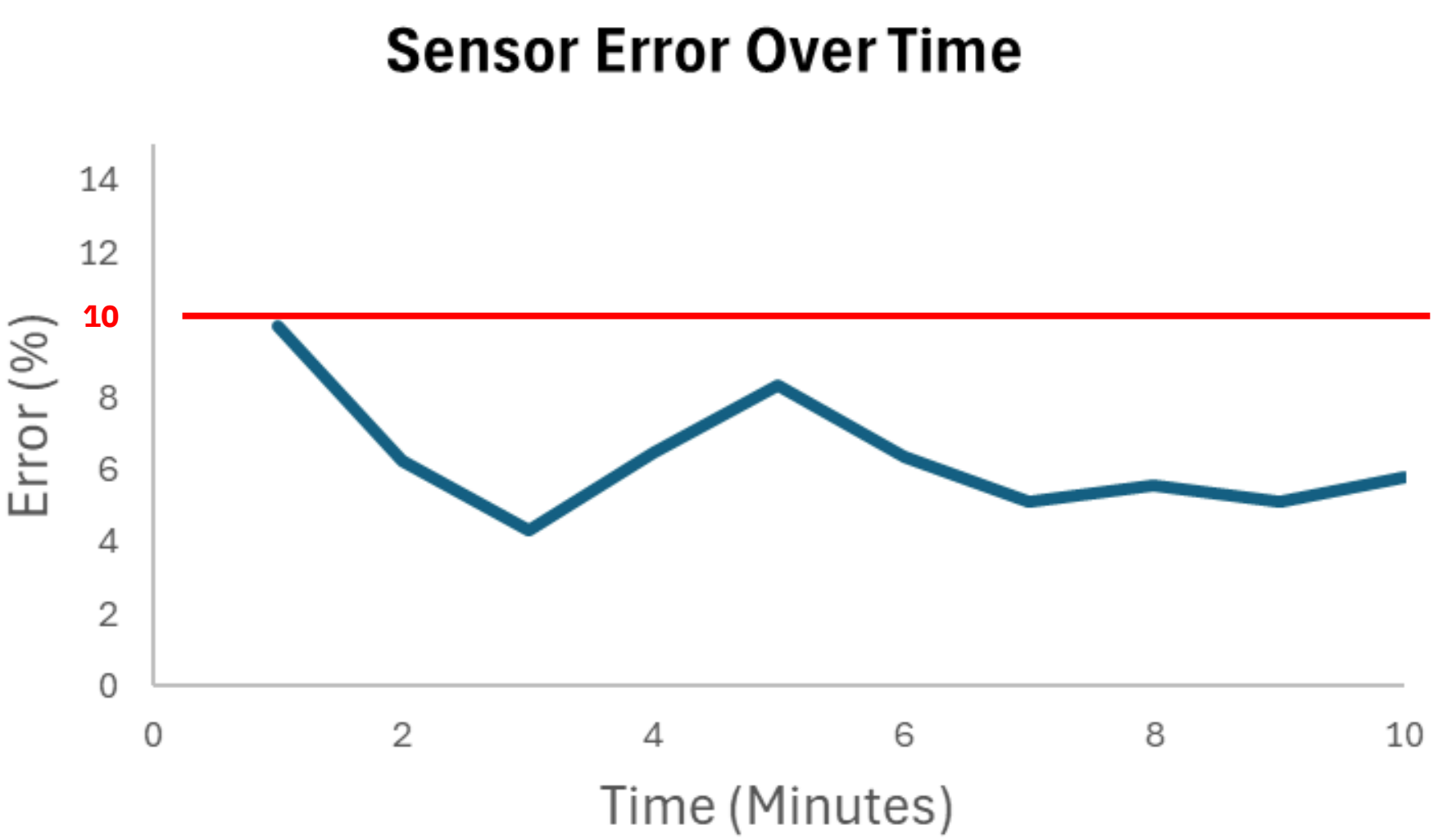


Thermal Output



Results

Updates once per **5 minutes**
< 10% error over 10 minutes
Stores **no** identifiable user data
Costs **\$180** per device
User intervention required **2x per day**
Only required at **entrances/exits**



Future Work

- UI + data upload implementation.
- Improve long-term robustness of accuracy.
- Accuracy and longevity testing.
- Rec Center installment and testing.



Acknowledgments

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